Data Science & Machine Learning Concepts

1. What is Data Wrangling?

Data Wrangling is the process of cleaning, organizing, and transforming raw data into a useful format for analysis.
It is also called data munging. Real-world data is often incomplete, inconsistent, or messy, so data wrangling prepares the data before analysis or machine learning.

Main steps:
• Collecting data from different sources
• Handling missing values
• Removing duplicates
• Correcting wrong formats
• Filtering unnecessary data
• Transforming data into proper structure

Example:
If a dataset contains missing ages, duplicate rows, and different date formats, data wrangling fixes all these issues before analysis.

2. What are NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn and their usage?

NumPy:
NumPy is a Python library used for numerical operations and array processing.
Usage:
• Fast mathematical calculations
• Matrix operations
• Scientific computing

Pandas:
Pandas is used for data analysis and manipulation.
Usage:
• Reading CSV/Excel files
• Handling missing values
• Filtering and grouping data

Matplotlib:

Matplotlib is used for creating charts and graphs.

Usage:

• Line charts

• Bar charts

• Pie charts

• Histograms

Seaborn:

Seaborn is a statistical visualization library built on Matplotlib.

Usage:

• Attractive statistical graphs

• Heatmaps

• Distribution plots

• Correlation analysis

Scikit-learn:

Scikit-learn is a machine learning library.

Usage:

• Classification

• Regression

• Clustering

• Model evaluation

3. If there is Matplotlib then why Seaborn is used?

Matplotlib provides basic plotting functionality, while Seaborn provides advanced and more attractive statistical visualizations.

Differences:

• Seaborn gives better default designs.

• Seaborn requires less code.

• Seaborn works very well with Pandas DataFrames.

• Seaborn is mainly used for statistical plots like heatmaps and pair plots.

Example:

Creating a heatmap in Matplotlib is difficult, but Seaborn provides a direct heatmap() function.

4. Different types of charts

Line Chart:

Used to show trends over time.

Bar Chart:

Used to compare categories.

Pie Chart:

Used to show percentage distribution.

Histogram:

Shows frequency distribution of continuous data.

Scatter Plot:

Shows relationship between two variables.

Heatmap:

Displays data using color intensity.

Box Plot:

Used to detect outliers and understand data spread.

5. What are Data Processing, Data Transformation, and Data Visualization?

Data Processing:

The process of collecting and preparing raw data into meaningful information.

Data Transformation:

Converting data from one format or structure into another.

Examples:

- Normalization
- Scaling
- Encoding categorical data

Data Visualization:

Representing data graphically to understand patterns and insights.

Examples:

• Charts
• Graphs
• Dashboards

6. What is Data Formatting?

Data Formatting means arranging data into a standard and readable format.

Examples:

• Formatting dates as DD/MM/YYYY
• Currency formatting
• Text capitalization
• Setting decimal precision

Importance:

• Improves readability
• Ensures consistency
• Helps accurate analysis

7. What are Mean, Median, Mode, and Standard Deviation with formulas?

Mean:

Average value of data.

Formula:

Mean = Sum of all observations / Number of observations

Median:

Middle value after arranging data in order.

Mode:

Most frequently occurring value.

Standard Deviation:

Measures spread of data around the mean.

Formula:

Standard Deviation = Square root of Variance

Low standard deviation means data is close to mean.

High standard deviation means data is widely spread.

8. What are Accuracy, Precision, Recall, and F1 Score with formulas?

Accuracy:

Percentage of correct predictions.

Formula:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision:

Out of predicted positives, how many are actually positive.

Formula:

Precision = TP / (TP + FP)

Recall:

Out of actual positives, how many are correctly identified.

Formula:

Recall = TP / (TP + FN)

F1 Score:

Harmonic mean of precision and recall.

Formula:

F1 Score = 2 × (Precision × Recall) / (Precision + Recall)

Where:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

9. What are First Quartile, Third Quartile, and IQR with formulas?

First Quartile (Q1):

25% of data lies below this value.

Third Quartile (Q3):

75% of data lies below this value.

Interquartile Range (IQR):

Difference between Q3 and Q1.

Formula:

IQR = Q3 – Q1

Usage:

IQR is used to detect outliers.

10. What is Data Normalization?

Data normalization scales data into a specific range, usually between 0 and 1.

Purpose:

- Improves machine learning performance
- Removes scale differences

Formula:

Normalized Value = (X – Min) / (Max – Min)

11. What are Label Encoding and One Hot Encoding?

Label Encoding:

Converts categories into numerical labels.

Example:

Red = 0, Blue = 1, Green = 2

One Hot Encoding:

Creates separate binary columns for each category.

Example:

Red → [1,0,0]

Blue → [0,1,0]

Difference:

Label encoding may create false order relationships, while one-hot encoding avoids this.

12. What are IQR and Z-score?

IQR:

Measures spread of middle 50% of data.

Formula:

IQR = Q3 – Q1

Z-score:

Measures how far a value is from the mean.

Formula:

Z = (X – Mean) / Standard Deviation

Usage:

Both are used for outlier detection.

13. What is Min-Max Scaler?

Min-Max Scaler scales data between 0 and 1.

Formula:

X_scaled = (X – Xmin) / (Xmax – Xmin)

Advantages:

- Keeps relationships between data
- Useful for neural networks

14. Formulas of Mean, Median, Min, Max, Variance, and Standard Deviation

Mean:

Mean = Sum of observations / Number of observations

Median:

Middle value after sorting.

Min:

Smallest value in dataset.

Max:

Largest value in dataset.

Variance:

Measures spread of data.

Formula:

Variance = Sum of squared differences from mean / Number of observations

Standard Deviation:

Square root of variance.

15. What is Heatmap?

A heatmap is a graphical representation of data where values are represented using colors.

Usage:

• Correlation analysis
• Pattern detection
• Visualizing matrix data

In machine learning, heatmaps are commonly used to show feature correlations.

16. What are RMSE and R2 Score for Boston Housing?

RMSE (Root Mean Square Error):

Measures prediction error in regression.

Formula:

RMSE = Square root of average squared prediction errors

Lower RMSE is better.

R2 Score:

Measures how well the model explains variance.

Formula:

$R^2 = 1 - $ (Residual Sum of Squares / Total Sum of Squares)

Higher R2 score is better.

R2 value close to 1 indicates good model performance.

17. Is higher or lower RMSE better? Same for R2 score.

RMSE:

Lower RMSE is better because it means prediction errors are small.

R2 Score:

Higher R2 score is better.

- R2 = 1 means perfect prediction.
- R2 = 0 means model explains no variance.
- Negative R2 means poor model.

18. What are Linear Regression and Logistic Regression and their usage?

Linear Regression:

Used to predict continuous values.

Examples:

- House price prediction
- Temperature prediction

Logistic Regression:

Used for classification problems.

Examples:

- Spam detection
- Disease prediction

Difference:

Linear regression gives numerical output, while logistic regression gives category/class output.

19. When to use Linear Regression and Logistic Regression and their difference?

Use Linear Regression when output is continuous.

Examples:

• Salary prediction

• Sales prediction

Use Logistic Regression when output is categorical.

Examples:

• Pass/Fail

• Yes/No

Main Difference:

Linear Regression:

• Output is continuous

• Uses straight line

Logistic Regression:

• Output is probability/class

• Uses sigmoid function

20. What is Classification and Confusion Matrix?

Classification:

A supervised learning task where data is classified into categories.

Examples:

• Spam or Not Spam

• Fraud or Not Fraud

Confusion Matrix:

A table used to evaluate classification performance.

It contains:

• TP (True Positive)

• TN (True Negative)

• FP (False Positive)

• FN (False Negative)

It helps calculate accuracy, precision, recall, and F1 score.

21. What is Standard Scaler?

Standard Scaler standardizes data by converting it into zero mean and unit variance.

Formula:

Z = (X – Mean) / Standard Deviation

Purpose:

• Makes features comparable

• Improves performance of ML algorithms

22. What are Tokenization, POS Tagging, Stopword Removal, Stemming, and Lemmatization?

Tokenization:

Breaking text into words or sentences.

POS Tagging:

Assigning grammatical labels like noun, verb, adjective.

Stopword Removal:

Removing common words like “is”, “the”, “and”.

Stemming:

Reducing words to root form.

Example:

Playing → Play

Lemmatization:

Converting words into meaningful root form.

Example:

Better → Good

These are common NLP preprocessing techniques.

23. What is TF-IDF?

TF-IDF stands for Term Frequency – Inverse Document Frequency.

It measures importance of a word in a document.

TF:

How often a word appears in a document.

IDF:

How unique the word is across documents.

Usage:

• Text classification

• Search engines

• NLP applications

24. What are Skewness and Kurtosis?

Skewness:

Measures asymmetry of data distribution.

• Positive skew: Tail on right side

• Negative skew: Tail on left side

Kurtosis:

Measures peakedness of distribution.

• High kurtosis: More outliers

• Low kurtosis: Fewer outliers

Both are used in statistical analysis.